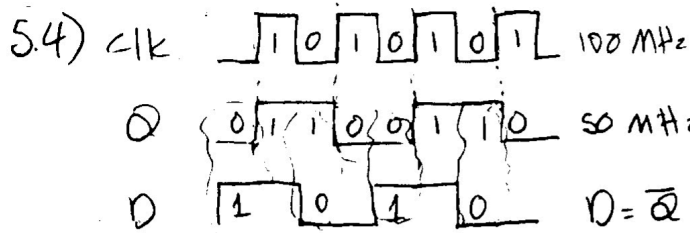
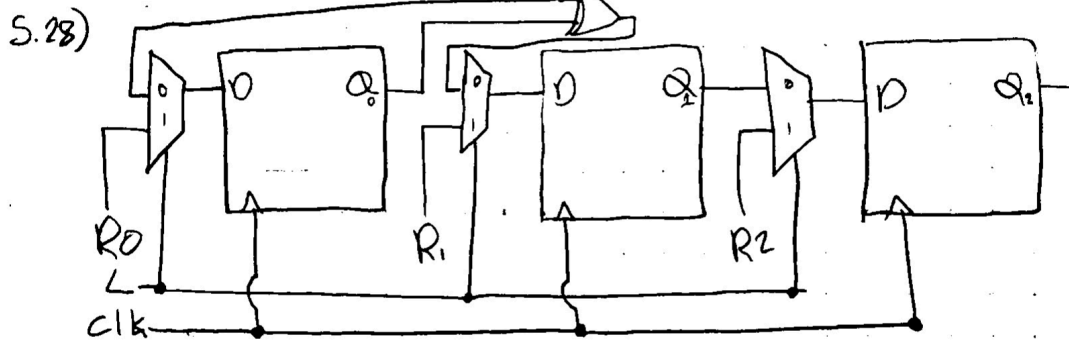
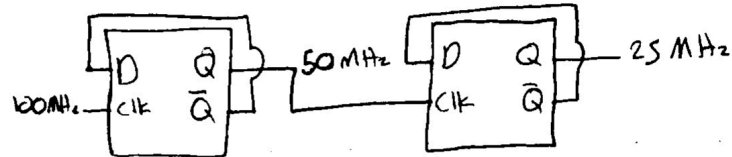


HW4

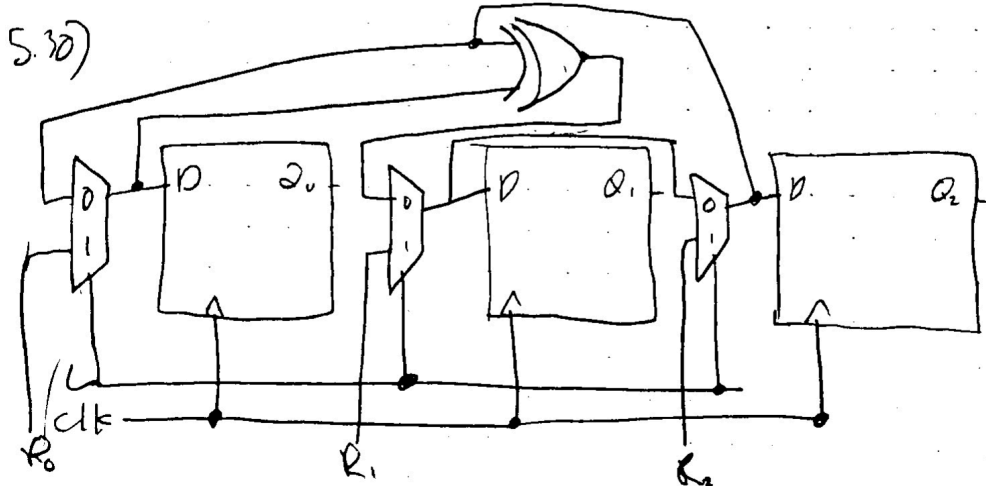
HW 4
10 4 26



clk	D	Q
0	0	X
0	1	X
1	0	1
1	1	0



$\{R_0, R_1, R_2\} = [0, 0, 1] \Rightarrow 1, 2, 4, 3, 6, 7, 5, \dots$ repeat.



No pattern, maybe race conditions?

5.10

```
module JK_FlipFlop(  
    input clk,  
    input clr,  
    input J,  
    input K,  
    output reg Q  
);  
    always @(posedge clk or posedge clr) begin  
        if (clr) begin  
            Q <= 1'b0;  
        end else begin  
            if (J & K)  
                Q <= ~Q;  
            if (J & ~K)  
                Q <= 1'b1;  
            if (~J & K)  
                Q <= 1'b0;  
            if (~J & ~K)  
                Q <= Q;  
        end  
    end  
endmodule
```

5.20

```
module modulo_12_counter (  
    input clk,  
    input reset,  
    input ena,  
    output reg [3:0] q  
);
```

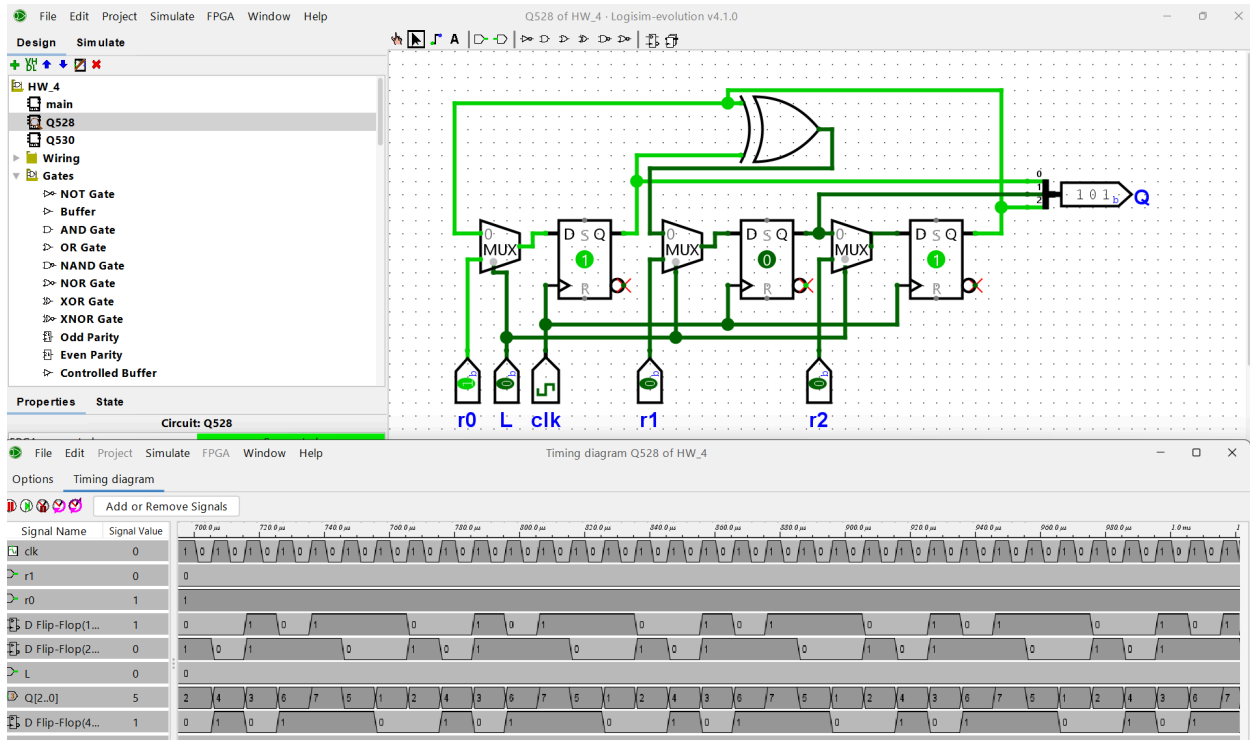
```

always @(posedge clk) begin
    if (reset) begin
        q <= 4'b0000;
    end else if (ena) begin
        if (q == 4'b1011)
            q <= 4'b0000;
        else
            q <= q + 4'b0001;
    end
end

endmodule

```

5.28



5.30

